

Introduction to Networks & OSI Model

1. Introduction to Networks

A computer network is a set of computers sharing resources located on or provided by network nodes.

- LAN (Local Area Network): Spans a small geographic area like a home or office.
- MAN (Metropolitan Area Network): Covers a city or a large campus.
- WAN (Wide Area Network): Covers a broad area, linking across metropolitan, regional, or national boundaries (e.g., the Internet).

2. Network Topologies

- Star: All nodes connect to a central hub/switch.
- Bus: All nodes share a single communication cable.
- Ring: Each node connects to exactly two other nodes, forming a ring.
- Mesh: Nodes are interconnected with many redundant interconnections.

3. OSI Model - 7 Layers

The Open Systems Interconnection (OSI) model standardizes the communication functions of a telecommunication or computing system.

Layer 7: Application Layer

Network process to application. Provides network services directly to the user's application (e.g., HTTP, FTP).

Layer 6: Presentation Layer

Data representation and encryption. Translates data between the application and network formats.

Layer 5: Session Layer

Interhost communication. Establishes, manages, and terminates connections between applications.

Layer 4: Transport Layer

End-to-end connections and reliability. Protocols include TCP and UDP.

Layer 3: Network Layer

Path determination and logical addressing (IP routing).

Layer 2: Data Link Layer

Physical addressing (MAC). Handles framing and error detection.

Layer 1: Physical Layer

Media, signal, and binary transmission. Transmits raw bit stream over the physical medium.